

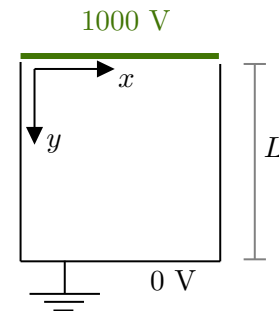
Project 4: Partial differential equations (PDEs)

Go to [Stud.IP](#) ► [Übung: Methods of Computational Physics](#) ► [CloCked](#) to upload a single PDF document with your results and answers by January 8th, 10am.

1. Laplace equation (10 points)

Consider the two-dimensional setup of a square box with a side length of L , where the top edge is set to a potential of $\phi(x, 0) = \phi_{\text{top}} = 1000 \text{ V}$ and the other three edges are grounded, i.e. at a potential of zero. Your task is to find the potential within the box, where it solves the Laplace equation

$$\Delta\phi(x, y) = 0, \quad (1)$$



where the x -axis is along the top edge and the y -axis points down, with the origin at the top left corner, as indicated in the sketch.

- a) Write a program that computes the potential $\phi(x, y)$ within the box with a discretisation length of $\Delta x = L/100$. Implement the Jacobi, Gauß-Seidel and SOR methods. As a stopping criterion, assert that the error of the discretised Laplace equation is smaller than $\epsilon_{\text{max}} = 10^{-3} \text{ V}$ everywhere. The error is defined as $\epsilon_{i,j} = |\phi_{i,j} - (\phi_{i+1,j} + \phi_{i-1,j} + \phi_{i,j+1} + \phi_{i,j-1})/4|$. Plot the average and maximum error versus the number of iterations for all algorithms. For SOR, use six different over-relaxation parameter values $\alpha = 0.5, 1.0, 1.25, 1.5, 1.75$ and 1.99 . Check if it still converges for $\alpha \geq 2.0$. Discuss your results. (6 points)

- b) The solution of this problem can be written as an infinite series:

$$\phi(x, y) = \sum_{n=1,3,5,\dots}^{\infty} \frac{4\phi_{\text{top}}}{n\pi} \sin\left(\frac{n\pi x}{L}\right) \frac{\sinh(n\pi(1 - y/L))}{\sinh(n\pi)}. \quad (2)$$

The sinh terms become very large for large n , which might cause large numerical errors in the ratio, or even overflows. Therefore it might be advantageous to use

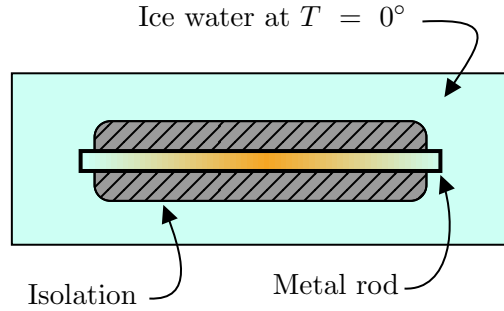
$$\frac{\sinh(n\pi(1 - y/L))}{\sinh(n\pi)} \rightarrow \exp(-n\pi y/L) \quad (3)$$

for larger n .

Plot the solution for 1, 10, 100 and 1000 terms. Plot the difference between one of the iterative solutions and the “infinite” series solution using 1000 terms. Discuss your results. (4 points)

2. Diffusion (16 points)

Consider a metallic rod of a finite length L and a small radius, which is isolated at its side, but not at its ends, where it is placed in contact with ice water at 0°C :



To simulate the temperature flow, we will need to solve the diffusion PDE

$$\frac{\partial T(x, t)}{\partial t} = \frac{K}{C\rho} \frac{\partial^2 T(x, t)}{\partial x^2}, \quad (4)$$

with the thermal conductivity K , the heat capacity C and the density ρ . Use $L = 1$, $K = 200$, $C = 1000$ and $\rho = 3000$, and, if not otherwise specified, $\Delta x = 0.01$ and $\Delta t = 0.1$ with a number of time steps $N_t = 10000$. The rod's temperature distribution is initially set to $T(x, 0) = \sin(\pi x/L)$.

- a) Simulate the system using the FTCS algorithm and plot $T(x, t)$. (8 points)
- b) The analytical solution is given by

$$T_{\text{exact}}(x, t) = \sin\left(\frac{\pi x}{L}\right) \exp\left(-\frac{\pi^2 K t}{L^2 C \rho}\right). \quad (5)$$

With that, we can define our simulation error as

$$\epsilon(t) = \frac{1}{N_x} \sum_{j=1}^{N_x-1} |T(j\Delta x, t) - T_{\text{exact}}(j\Delta x, t)|, \quad (6)$$

where $N_x = L/\Delta x$. Study the behaviour of $\epsilon(t = 10000)$ varying Δt between 0.001 and 1000 while keeping Δx fixed. What happens? Explain your finding. (4 points)

- c) Now, implement the implicit Euler backward, the Crank–Nicolson and Dufort–Frankel algorithms and repeat the analysis for $\epsilon(t = 10000)$ for all of them. Discuss your results comparing the four algorithms, in particular with respect to the scaling behaviour of the error with Δt . (4 points)

3. Solitons (16 points)

Water waves in shallow, narrow channels can be described by the Koorteweg–de Vries (KdV) equation

$$\frac{\partial u(x, t)}{\partial t} + \epsilon u(x, t) \frac{\partial u(x, t)}{\partial x} + \mu \frac{\partial^3 u(x, t)}{\partial x^3} = 0. \quad (7)$$

The non-linear term leads to a sharpening of the wave and ultimately to a shock wave. In contrast, the $\partial^3 u / \partial x^3$ term produces broadening. For the proper parameters and initial conditions, the two effects exactly balance each other, and a stable wave is formed, which is called a “soliton”. These stable solitons almost behave as particles, and appear in many areas of physics. For more details on the numerical discovery of this fascinating phenomenon, see “[Computer Simulations Led to Discovery of Solitons](#)”. For a deeper dive, you can also download the original paper from [Stud.IP ▶ Übung: Methods of Computational Physics ▶ Files ▶ Project 4: Partial differential equations](#).

Inserting the traveling wave ansatz $u(x, t) = u(x - ct)$ gives a solvable ODE with the (non-trivial) solution

$$u(x, t) = \frac{-c}{2} \operatorname{sech}^2 \left(\frac{1}{2} \sqrt{c} (\xi - \xi_0) \right), \quad (8)$$

where $\xi = x - ct$ is the phase. Note that the amplitude is proportional to the propagation speed c . Discretising the KdV equation gives the algorithm

$$\begin{aligned} u_j^{n+1} \approx & u_j^{n-1} - \frac{\epsilon}{3} \frac{\Delta t}{\Delta x} [u_{j+1}^n + u_j^n + u_{j-1}^n] [u_{j+1}^n - u_{j-1}^n] \\ & - \mu \frac{\Delta t}{\Delta x^3} [u_{j+2}^n + 2u_{j-1}^n - 2u_{j+1}^n - u_{j-2}^n]. \end{aligned} \quad (9)$$

To calculate u_j^2 , the first term on the right-hand side is not yet available, so we use a simpler forward difference scheme for the time step, which gives

$$\begin{aligned} u_j^2 \approx & u_j^1 - \frac{\epsilon}{6} \frac{\Delta t}{\Delta x} [u_{j+1}^1 + u_j^1 + u_{j-1}^1] [u_{j+1}^1 - u_{j-1}^1] \\ & - \frac{\mu}{2} \frac{\Delta t}{\Delta x^3} [u_{j+2}^1 + 2u_{j-1}^1 - 2u_{j+1}^1 - u_{j-2}^1]. \end{aligned} \quad (10)$$

In addition to the first time step, we also need to treat u_1^n , u_2^n , $u_{N_{\max}-1}^n$ and $u_{N_{\max}}^n$ separately, since also they can not be updated using Eq. (9). Given the initial state we will use below, a simple method is to use the constant boundary values $u_1^n = 1$ and $u_{N_{\max}}^n = 0$. This leaves u_2^n and $u_{N_{\max}-1}^n$. These are still to be calculated using Eqs. (9) and (10), but with the replacements $u_0^n \rightarrow u_1^n$ (when calculating u_2^n) and $u_{N_{\max}+1}^n \rightarrow u_{N_{\max}}^n$ (when calculating $u_{N_{\max}-1}^n$) on the right-hand sides.

- a) Derive Eq. (9). For the first derivatives, use the central difference approximations $\partial u / \partial t \approx (u_j^{n+1} - u_j^{n-1}) / (2\Delta t)$ and $\partial u / \partial x \approx (u_{j+1}^n - u_{j-1}^n) / (2\Delta x)$. To find an approximation for the third derivative expand $u(x, t)$ to including $\mathcal{O}(\Delta x^4)$ about the four points $u(x \pm \Delta x, t)$ and $u(x \pm 2\Delta x, t)$, then solve for $\partial^3 u / \partial x^3$. Finally, for the non-differentiated $u(x, t)$ in the second term of Eq. (9), use the average over three adjacent points, i.e. $u = (u_{j-1}^n + u_j^n + u_{j+1}^n) / 3$. In your derivation, keep track of the truncation error, to prove that Eq. (9) is of order $\mathcal{O}(\Delta t^3) + \mathcal{O}(\Delta t \Delta x^2)$. (4 points)

b) Implement the above algorithm solving the KdV equation for the initial condition

$$u(x, t = 0) = \frac{1}{2} \left[1 - \tanh \left(\frac{x - 25}{5} \right) \right]. \quad (11)$$

The parameters are given by $\epsilon = 0.2$ and $\mu = 0.1$ and the step sizes by $\Delta x = 0.4$ and $\Delta t = 0.1$. Use 130 steps in x , such that our simulation volume is given by $L = 130\Delta x = 52$. Verify that these constants satisfy the stability condition

$$\frac{\Delta t}{\Delta x} \left[\epsilon |u| + 4 \frac{\mu}{(\Delta x)^2} \right] \leq 1. \quad (12)$$

Store the solution every 250 or so time steps and run the simulation for about 2000 time steps. Plot the disturbance u versus position and versus time. Scott Russell observed in 1834 in the Edinburgh-Glasgow canal that an initial, arbitrary wave form set in motion evolves into two or more solitary waves that move at different velocities. Can you confirm his observation? Into how many solitary waves does your initial wave form break up into? (8 points)

c) Explore what happens when a tall soliton collides with a short one. Do they bounce off each other? Do they go through each other? Do they interfere? Do they destroy each other? Does the tall soliton still move faster than the short one after collision? Start off by placing a tall soliton of height 0.8 at $x = 12$, and a small soliton in front of it at $x = 26$:

$$u(x, t = 0) = 0.8 \left[1 - \tanh^2 \left(\frac{3x}{12} - 3 \right) \right] + 0.3 \left[1 - \tanh^2 \left(\frac{4.5x}{26} - 4.5 \right) \right]. \quad (13)$$

Make sure to adapt the constant boundary values for u_1^n and $u_{N_{\max}}^n$ for this simulation. (4 points)